

WANYUAN, China

WMO: 572370

Lat: 32.0744N

Lon: 108.0317E

Elev: 2211

StdP: 13.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.3	11.9	11.0	39.4	16.1	13.4	37.7	23.6	41.5	19.6	40.4	2.9	320	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	93.4	74.4	91.2	73.9	89.0	73.1	77.1	88.3	76.2	86.4	75.3	84.6	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	140.3	80.7	73.6	136.0	79.9	72.7	131.9	79.4	42.5	88.1	41.5	87.1	40.6	84.6	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	16.0	13.0	23.7	96.9	2.9	1.9	21.6	98.3	19.9	99.4	18.3	100.5	16.2	101.9
			21.8	79.0	2.7	1.2	19.8	79.8	18.2	80.5	16.7	81.2	14.7	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	40.2	44.3	52.5	61.8	68.1	74.2	79.1	78.5	70.3	61.2	51.9	42.9
	DBStd	14.28	4.43	5.00	6.60	6.28	4.80	4.34	3.84	4.88	5.25	5.16	5.48	4.10
	HDD50	794	304	169	51	4	0	0	0	0	0	1	44	221
	HDD65	3123	769	580	390	131	25	1	0	0	10	140	393	684
	CDD50	4627	1	9	127	357	562	727	902	883	609	347	101	2
	CDD65	1481	0	0	2	34	121	278	437	419	169	21	0	0
	CDH74	12264	0	0	41	374	895	2062	3923	3805	1038	125	1	0
	CDH80	4722	0	0	6	87	242	726	1642	1666	338	15	0	0
Wind	WSAvg	4.1	4.2	4.6	4.9	4.8	4.5	3.6	3.3	3.7	3.6	3.6	3.8	3.9
Precipitation	PrecAvg	48.7	0.2	0.4	1.5	3.5	5.4	5.6	9.8	7.5	8.9	4.1	1.5	0.3
	PrecMax	87.3	0.7	1.7	4.2	8.1	10.4	10.6	23.5	22.4	20.9	9.3	4.1	1.6
	PrecMin	30.4	0.0	0.0	0.3	0.7	1.3	1.1	3.6	1.4	0.5	1.0	0.4	0.0
	PrecStd	11.8	0.2	0.4	0.9	1.7	2.6	2.8	5.0	4.6	5.2	2.3	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	64.9	79.1	87.1	89.6	93.3	95.4	96.5	92.5	82.3	71.6	59.5
		MCWB	45.3	49.9	59.5	64.0	66.7	71.7	76.1	75.2	72.7	65.8	57.4	47.7
	2%	DB	54.2	60.0	73.1	82.4	85.7	89.9	92.9	93.9	87.6	77.7	67.5	55.7
		MCWB	43.5	47.5	56.2	62.1	66.3	71.6	75.2	74.7	70.7	63.1	55.3	45.5
	5%	DB	50.7	56.4	68.7	78.3	82.5	87.1	90.8	91.5	83.8	73.8	63.8	52.9
		MCWB	41.7	45.9	53.7	60.5	65.2	70.7	74.9	74.1	69.2	61.7	54.1	44.5
	10%	DB	48.0	52.9	64.4	74.2	79.1	84.4	88.6	89.1	79.9	69.9	60.5	50.2
		MCWB	40.7	44.4	51.9	59.4	64.4	70.2	74.5	73.7	68.0	60.4	52.6	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	51.8	62.1	68.2	71.2	76.2	78.9	78.2	75.4	68.0	59.5	50.3
		MCDB	55.3	60.6	73.7	81.9	80.8	86.3	89.9	90.9	87.0	77.1	66.9	55.9
	2%	WB	45.0	49.3	57.6	65.0	69.1	74.6	77.4	76.9	73.6	65.4	57.4	47.7
		MCDB	51.6	56.7	68.9	76.6	79.0	83.7	88.0	88.9	82.3	72.6	64.1	53.0
	5%	WB	43.0	47.4	55.1	62.6	67.5	73.3	76.5	75.9	71.7	63.7	55.8	45.8
		MCDB	49.5	54.1	66.3	74.1	77.3	82.1	86.4	86.9	78.8	70.3	61.3	51.1
	10%	WB	41.2	45.6	53.0	60.6	66.2	71.9	75.7	75.0	69.9	62.0	54.3	44.2
		MCDB	47.2	52.1	62.8	71.1	75.8	80.2	84.7	85.1	76.8	68.3	59.1	49.0
Mean Daily Temperature Range	5% DB	MDBR	12.4	13.3	16.4	17.8	16.9	15.9	15.9	16.7	14.3	13.4	13.4	12.6
		MCDBR	19.9	21.8	25.8	27.2	25.0	22.4	20.6	21.5	21.9	21.5	21.1	19.3
		MCWBR	11.0	11.3	10.8	9.7	8.5	6.9	5.7	5.7	7.1	8.2	9.9	10.5
	5% WB	MCDBR	17.5	17.7	22.9	22.9	19.3	17.2	17.3	18.6	17.2	16.9	16.5	15.8
		MCWBR	10.2	9.9	10.0	9.5	7.8	6.3	5.3	5.6	6.6	7.3	8.4	9.1
Clear-Sky Solar Irradiance	taub	0.611	0.665	0.678	0.649	0.599	0.584	0.589	0.610	0.599	0.606	0.626	0.586	
	taud	1.569	1.504	1.521	1.616	1.773	1.856	1.878	1.806	1.792	1.698	1.606	1.631	
	Ebn at Noon	181	188	201	217	230	233	231	224	218	201	176	178	
	Edh at Noon	71	83	88	82	71	65	63	67	66	67	68	64	
All-Sky Solar Radiation	RadAvg	597	665	1018	1299	1347	1419	1531	1420	1012	801	626	548	
	RadStd	101	132	102	160	181	158	232	177	141	146	111	99	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		+0.63	N/A	N/A	+1.22	N/A	N/A	-68	-155	+175	+74

Nomenclature: See separate page